Latent World Action Model for GUI Agents

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Paper version: arXiv 2610.14228v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *explicitly reconstructing future screens introduces an expensive and unnecessarily broad intermediate task into action selection*

Render a screen, re-perceive it — or **predict the consequence in latent space**.

Why not render the future screen

1. **L1 cost.** Rendering a future screen is slow; a scorer re-reads it.
2. **L2 waste.** Action choice needs the consequence, not pixels.
3. **Shortcut.** Pixel predictions copy the screen on big transitions.

Evaluation setup

AndroidWorld · 116 tasks | AndroidControl (offline) | GUI-Odyssey (OOD) | 5 agent backbones | K = 3 candidates | judge: Qwen3-VL-8B LoRA | seed 30 · 1 run

- **Online:** AndroidWorld; same candidates for every method.
- **Offline:** step Type / Grounding / SR; no variance.

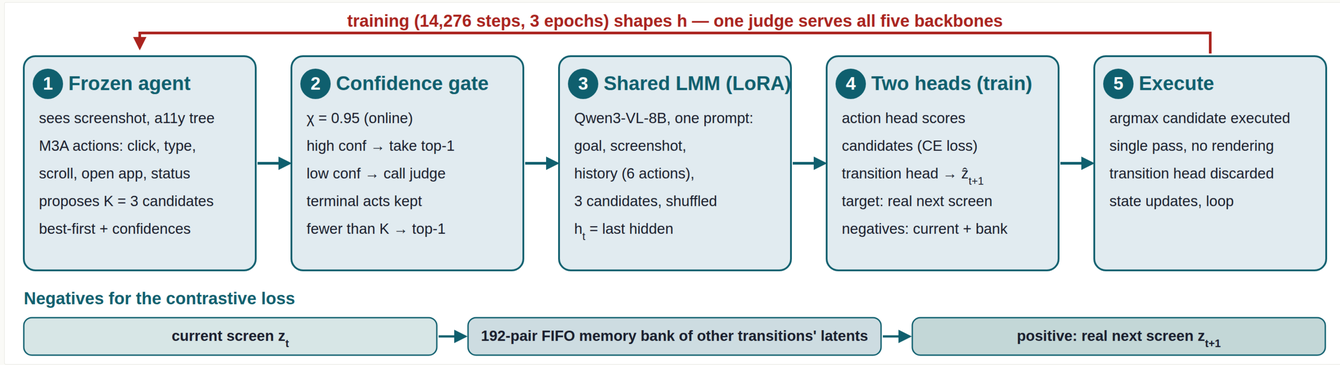
Our read (CUA Radar)

62	★★	online
score of 100	band	evidence level

- **Reports:** online eval; OOD benchmark; cost; ablation; matched baselines; code.
- **Flags:** no variance; budget match partial; judge unvalidated.
- Pattern K02. Authors' numbers, not rerun.

Method: a latent world model that scores candidates

- **Sees / acts:** screenshot + a11y tree; M3A actions; frozen agent proposes K = 3 candidates.
- **Signal:** cross-entropy on the annotated action + contrastive next-state latent (InfoNCE).
- **Deploy:** gate at χ = 0.95; one forward pass; transition head discarded.



Our redraw (not the paper's figure): the gated loop and the two training heads.

$$L = \mathcal{L}_{\text{action}} + \lambda \mathcal{L}_{\text{transition}}$$

joint loss (λ = 1.0, τ = 0.07, LoRA r = 16)

Key results: AndroidWorld online SR (%)

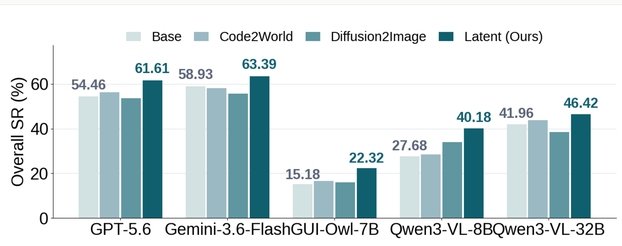


Table 2, redrawn: Overall SR, one run per method, same candidates.

+4.5 to +12.5 pts over base agents; pixel world models mixed.

SO WHAT → *Offline step SR kept apart: ours +4.8, pixel WMs −8.6.*

Ablation: AndroidWorld SR (%)

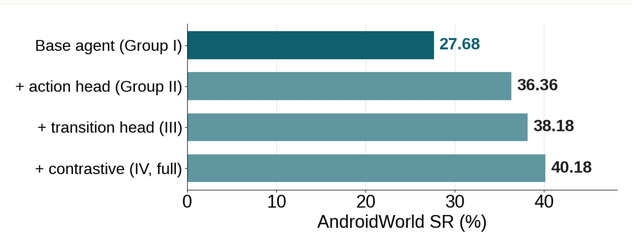


Table 4, redrawn: Qwen3-VL-8B backbone, one component added at a time.

SO WHAT → *Paper: 27.68 → 36.36 (action head); +1.82 transition; +2.00 contrastive; +12.50 total over the base agent. Group II is pure latent rescoring before any world-model supervision; Group III trains the transition head with point-wise cosine regression; Group IV swaps in the action-aware contrastive objective. All variants stay in latent space, no future-screen generation.*

Headline Numbers

40.18%
AndroidWorld SR
vs 27.68 base

- **+12.50** pts over base
- **98.9%** less WM latency
- **63.39%** top SR (Gemini-3.6)

The paper's own controls

CONTROL	LATENT (OURS)	BASELINES
Candidates	same K = 3 sets	all world models
Offline SR	up to +4.8 vs base	pixel WMs: −8.6
OOD GUI-Odyssey	34.73 / 52.13	base 33.73 / 52.07
Run time, GPT-5.6	141.4 s	base 127.4, C2W 370.9

Takeaway

Score consequences, not pixels: highest AndroidWorld SR, all five backbones.

+12.50 pts (ablation) | 98.9% latency cut | 43.6M LoRA params | code released

Scan to Read

Paper | Code